

Understanding Primary Keys & Foreign Keys

Step 1: Create the **departments** table

None

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50)  
);
```

Step 2: Insert data into **departments**

None

```
INSERT INTO departments VALUES  
(101, 'Human Resources'),  
(102, 'Information Technology'),  
(103, 'Sales'),  
(104, 'Finance');
```

Step 3: Create the **employees** table

None

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(50),  
    age INT,  
    salary DECIMAL(10,2),  
    department_id INT,  
    FOREIGN KEY (department_id) REFERENCES  
departments(department_id)  
);
```

Step 4: Insert data into **employees**

None

```
INSERT INTO employees VALUES
```

```
(1, 'Aarav', 25, 45000, 101),  
(2, 'Diya', 28, 60000, 102),  
(3, 'Rahul', 30, 55000, 103),  
(4, 'Meera', 27, 70000, 102),  
(5, 'Kiran', 35, 80000, 104);
```

Practice Questions

1. Display all departments.

None

```
SELECT * FROM departments;
```

2. Display all employees.

None

```
SELECT * FROM employees;
```

3. Show employee name and department ID.

None

```
SELECT employee_name, department_id  
FROM employees;
```

4. Display employees working in department 102.

None

```
SELECT *  
FROM employees  
WHERE department_id = 102;
```

5. Find employees with a salary greater than 60,000.

None

```
SELECT *  
FROM employees  
WHERE salary > 60000;
```

6. Display employee names in ascending order.

None

```
SELECT *  
FROM employees  
ORDER BY employee_name;
```

Test the Constraints

A. Try inserting a duplicate Primary Key.

None

```
INSERT INTO departments VALUES  
(101, 'Marketing');
```

Expected: ❌ Error because `department_id` is a **Primary Key** and must be unique.

B. Try inserting a NULL Primary Key.

None

```
INSERT INTO departments VALUES  
(NULL, 'Marketing');
```

Expected: ❌ Error because a **Primary Key** cannot be `NULL`.

C. Try inserting an employee with a non-existing department.

None

```
INSERT INTO employees VALUES  
(6, 'John', 29, 50000, 999);
```

Expected: ❌ Error because `999` does not exist in the `departments` table. This violates the **Foreign Key** constraint.

D. Try deleting a department that is being referenced.

None

```
DELETE FROM departments  
WHERE department_id = 102;
```

Expected:  Error because employees are already assigned to department 102.